

Towards ISP-Free Fisheye Camera Perception: An Evaluation in Image and BEV Space

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Abstract

Conventional autonomous perception pipelines rely heavily on ISP-processed RGB imagery, which is optimized for human visual aesthetics but frequently discards radiometric data critical for machine cognition. While prior RAW-domain research has demonstrated the benefits of sensor-domain learning, these studies predominantly address 2D image-space detection utilizing narrow-field rectilinear cameras. Modern autonomous driving systems, however, increasingly demand 3D Bird’s-Eye-View (BEV) spatial aggregation derived from complex surround-view fisheye cameras. This work systematically investigates ISP-free perception directly in the sensor domain using un-demosaiced 16-bit Bayer RAW automotive fisheye imagery. To support this investigation, we utilize the RAW-Det-Fisheye dataset, comprising 168,000 synchronized, multi-camera frames containing aligned 16-bit sensor RAW and ISP-processed RGB representations. To comprehensively evaluate sensor-domain resilience, we analyze performance across both 2D image-space detection using YOLOv8 and 3D BEV-space reasoning using BEVFormer equipped with multi-view feature aggregation. Experimental evaluations reveal that despite the severe non-linear radial distortion of fisheye lenses and the inherent sparsity of Bayer patterns, sensor-domain RAW representations achieve semantic detection and spatial localization performance comparable to conventional ISP-processed RGB baselines. These findings conclusively demonstrate that robust geometric consistency and semantic features are preserved prior to ISP intervention, motivating the development of highly efficient, sensor-native perception architectures tailored explicitly for machine vision.

Keywords: Fisheye Perception, Image Signal Processor Free Detection, Sensor-Native Learning, BEV Perception, Autonomous Driving, Raw Fisheye Dataset

1 Introduction

Modern autonomous driving systems increasingly rely on automotive fisheye cameras due to their ultra-wide field of view, enabling surround-view perception, near-field obstacle detection, and automated parking functionalities [Kumar et al., 2022]. However, fisheye imaging introduces severe radial distortion and spatially varying resolution, creating significant challenges for downstream perception algorithms. The measured RAW signal is

$$R_c(x, y) = g \cdot t \cdot I_c(x, y) + n_{shot} + n_{read} \quad (1)$$

where g is sensor gain, t is exposure time, and n_{shot} and n_{read} denote shot and read noise. Most automotive cameras employ a Bayer Color Filter Array (CFA), producing a sparse single-channel RAW representation. Conventional pipelines convert this sensor-domain data into RGB imagery through an Image Signal Processor (ISP):

$$I_{RGB} = f_{ISP}(R) \quad (2)$$

where f_{ISP} includes demosaicing, white balance, gamma correction, tone mapping, and denoising.

Although ISP processing improves visual quality for human observers, it may alter the radiometric fidelity of sensor measurements [Brooks et al., 2018, Li et al., 2024a]. Consequently, recent work has explored RAW-domain learning for image restoration and image-space perception [Chen et al., 2018, Xu et al., 2023, Cui and Harada, 2024, Morawski et al., 2022, Guo et al., 2025, Li et al., 2024a].

However, prior studies have largely focused on 2D image-space reasoning using conventional cameras. The effectiveness of sensor-domain RAW representations for Bird’s-Eye-View (BEV) perception under fisheye imaging remains largely unexplored. To address this gap, we investigate ISP-free autonomous perception using 16-bit Bayer RAW automotive fisheye imagery. Using the proposed RAW-Det-Fisheye dataset, YOLOv8 and BEVFormer, we evaluate whether semantic and geometric representations can be learned directly from sensor measurements.

2 Contributions

The primary contributions of this work are summarized as follows:

1. **ISP-Free Fisheye Perception Paradigm:** We pioneer an ISP-free perception pipeline tailored specifically for complex automotive fisheye cameras. We demonstrate that machine cognition can operate directly on mosaiced, 16-bit Bayer RAW sensor measurements, effectively bypassing the radiometric information loss and hallucination artifacts inherent in human-centric ISP pipelines.
2. **Dual-Space Architectural Evaluation:** We provide the first systematic evaluation of sensor-domain robustness across two distinct spatial paradigms: 2D image-space reasoning (utilizing YOLOv8) and 3D multi-view Bird’s-Eye-View (BEV) perception (utilizing BEVFormer). This effectively addresses both the classical camera-based and modern BEV-based perception applications. To our knowledge, this is the first work to evaluate the ISP influence on BEV tasks.
3. **RAW-Det-Fisheye Dataset:** We evaluated the models on a comprehensive, large-scale industry grade automotive dataset comprising 168,000 synchronized frames of aligned 16-bit sensor RAW and ISP-processed RGB representations. By maintaining identical ground-truth bounding box annotations across domains, this dataset enables tightly controlled, rigorous evaluation of sensor-domain learning against traditional imaging pipelines.
4. **Demonstration of Geometric Resilience:** Our empirical findings establish that multi-view spatial aggregation and 3D geometric projection can successfully occur prior to ISP intervention. We prove that sensor-domain RAW representations preserve sufficient semantic structure and geometric consistency to achieve detection and localization performance comparable to conventionally optimized RGB baselines.

3 State of the Art

3.1 RAW-Domain Learning and Neural ISPs

Traditional Image Signal Processors (ISPs) rely on handcrafted, sequential modules designed to produce visually appealing RGB images for human consumption. However, this human-centric optimization often destroys low-level radiometric details essential for machine cognition. Early RAW-domain research addressed this by developing deep learning models for extreme low-light imaging [Chen et al., 2018] and introducing inverse ISP modeling to recover sensor data from sRGB images [Brooks et al., 2018]. Subsequent works explored mapping RAW signals to sRGB domains using neural networks [Zhang et al., 2021] or explicitly optimizing

Neural ISPs for downstream machine perception tasks [Morawski et al., 2022]. More recently, approaches like Dark-ISP [Guo et al., 2025] have further enhanced RAW image processing specifically to boost low-light object detection. While these methods demonstrate the value of preserving RAW data, they generally still rely on an intermediate processing step to reconstruct a dense representation before applying perception networks, adding unnecessary computational overhead.

3.2 RAW-Domain Perception in Image Space

To eliminate the computational bottleneck of explicit ISP reconstruction, recent literature has shifted toward direct ISP-free perception. Foundational works have established new benchmarks and models specifically for 2D object detection directly on raw sensor measurements [Xu et al., 2023, Li et al., 2024b]. To leverage the vast knowledge in models pre-trained on RGB data, techniques like RAW-Adapter [Cui and Harada, 2024] have been introduced to efficiently adapt standard visual models to camera RAW inputs without full retraining. Furthermore, direct processing of un-demosaiced snapshots has been shown to significantly improve overall visual computing efficiency [Li et al., 2024a]. These studies collectively prove that neural networks can extract robust semantic features directly from sparse Bayer-patterned data.

3.3 Bridging the Gap: From 2D to BEV-Space Fisheye Perception

Despite the success of RAW-domain learning in 2D image space, modern autonomous driving pipelines increasingly depend on 3D Bird’s-Eye-View (BEV) perception architectures. BEV perception requires rigorous geometric feature projection, multi-view spatial aggregation, and 3D reasoning. Performing these complex operations directly on sparse, un-demosaiced Bayer data presents a fundamentally different challenge than standard 2D detection, particularly when utilizing automotive fisheye cameras characterized by extreme non-linear radial distortion.

Paradigm	Processing Pipeline	Limitations Addressed by Our Work
Traditional HW ISP	RAW $\rightarrow$ HW ISP $\rightarrow$ RGB $\rightarrow$ Detect	Discards linear radiometric data; demosaicing hallucinates edges; optimized for humans, not machines.
Neural / Task-ISP	RAW $\rightarrow$ Neural ISP $\rightarrow$ RGB $\rightarrow$ Detect	Retains dependency on dense intermediate reconstruction, leading to high latency.
2D RAW Detection	RAW $\rightarrow$ 2D Object Detection	Lacks 3D geometric reasoning; primarily evaluated on rectilinear, narrow-field lenses.
Proposed Framework	Fisheye RAW $\rightarrow$ BEV/2D Detect	Investigates direct 3D BEV spatial aggregation under extreme fisheye distortion without demosaicing.

Table 1: Comparison of traditional, neural ISP, and sensor-direct perception pipelines.

As summarized in Table 1, prior research has predominantly focused on rectilinear 2D tasks. This work explicitly targets the unexplored intersection of ISP-free perception, fisheye geometry, and 3D BEV spatial aggregation. By demonstrating that sensor-domain RAW representations remain geometrically stable during complex BEV transformations, our research bridges the critical gap between raw sensor physics and state-of-the-art autonomous driving architectures.

4 The RAW-Det-Fisheye Dataset

To rigorously evaluate the efficacy of ISP-free perception in autonomous driving, we evaluate using the **RAW-Det-Fisheye** dataset. Existing autonomous driving benchmarks predominantly provide heavily compressed, ISP-processed RGB imagery, while established RAW-domain datasets are largely restricted to narrow-field,

rectilinear cameras. This dataset bridges that critical gap by providing large-scale, synchronized multi-camera fisheye imagery specifically curated for sensor-domain machine perception.

4.1 Dataset Composition and Sensor Characteristics

The dataset comprises a total of 168,000 frames, carefully partitioned into 160,000 training frames and 8,000 validation frames. The imagery is captured using automotive-grade surround-view fisheye cameras characterized by extreme non-linear radial distortion Figure 1. The dataset was collected directly from automotive image sensors in RAW Bayer format prior to ISP processing. The Operational Design Domain includes urban, suburban, parking and highway driving environments containing vehicles, pedestrians, cyclists, road infrastructure, and traffic control devices. Data were recorded under diverse illumination conditions ranging from bright daylight to low-light nighttime scenes to evaluate perception performance across a broad spectrum of sensor operating conditions. The dataset consists of 16-bit Bayer RAW images in BGGR format captured using a Dual Conversion Gain (DCG) HDR automotive fisheye sensor. Native sensor frames are acquired at a resolution of 1632×1290 pixels and subsequently downsampled to 1024×864 while preserving the Bayer mosaic structure. The downsampling operation is performed directly on the RAW data without demosaicing or color interpolation, ensuring that the original CFA sampling pattern is maintained.

Data was collected at a frequency of 10 Hz over 50 hours of recordings. The data is geographically diverse, ensuring the model’s robustness across different regional infrastructures, with the distribution: **Czech Republic: 44%, UK & Republic of Ireland: 22%, USA: 17%, Germany: 13%, Denmark: 1.5%, Others: 2.5%.**

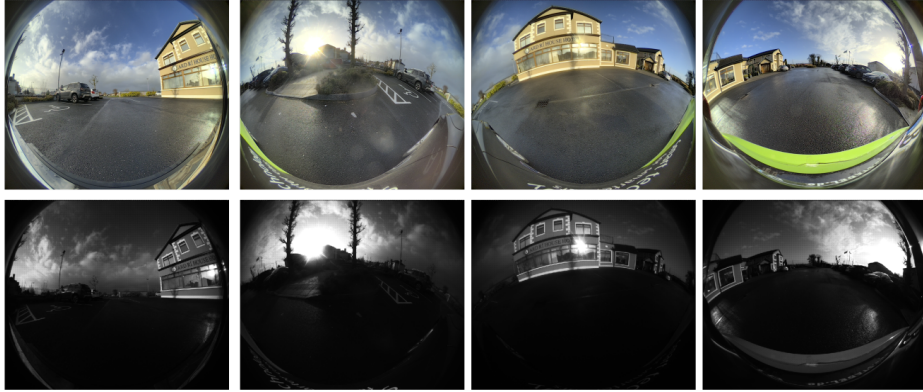


Figure 1: ISP Processed and RAW Bayer Fisheye images from our Dataset. Raw images are enhanced for visualization purposes.

More samples and quality results will be shared as a Supplementary material and will be uploaded to the arxiv

4.2 Multi-Domain Aligned Representations

A core feature of the RAW-Det-Fisheye dataset is the strict temporal and spatial alignment of multiple imaging domains for every captured frame. This enables perfectly controlled comparisons between sensor-native processing and conventional pipelines. Each frame is provided in two distinct representations: (i) The direct, 16-bit linear radiometric output from the image sensor, preserving the original dynamic range and containing inherent physical noise (e.g., shot and read noise) prior to any ISP intervention (ii) The conventionally processed 32-bit, 3-channel sRGB output generated via the camera’s hardware ISP. This serves as the primary baseline representing human-optimized perception.

4.3 Annotation Consistency

To ensure that empirical performance variations are exclusively attributable to the input data representation rather than label discrepancies, ground-truth bounding box annotations are rigidly identical across both do-

mains. This strict alignment guarantees that subsequent evaluations in both 2D image-space and 3D BEV-space accurately reflect the semantic and geometric resilience of the underlying pixel representations.

5 Proposed Methodology

To rigorously evaluate the viability of ISP-free perception, we deploy two state-of-the-art architectures: YOLOv8 [Jocher et al., 2023] for 2D image-space reasoning and BEVFormer [Li et al., 2022] for 3D multi-view spatial aggregation. By isolating the perception models from the ISP, we can directly assess the neural network’s ability to interpret raw sensor physics.

5.1 Image-Space Perception Using YOLOv8

Image-space perception evaluates semantic understanding directly in 2D sensor coordinates. We utilize YOLOv8 [Jocher et al., 2023], a highly efficient single-stage object detector featuring a modified CSPDarknet backbone, a Path Aggregation Network (PANet) neck, and an anchor-free detection head. For our ISP-free baseline, YOLOv8 is trained directly on the single-channel Bayer mosaiced RAW input without demosaicing or any ISP preprocessing:

$$X_{\text{RAW}} \in \mathbb{R}^{H \times W \times 1} \quad (3)$$

To prevent the model from inheriting biases optimized for dense sRGB imagery, and to preserve the physical properties of the sensor data, we applied the following specific training configurations. The training was started from scratch with no pretrained RGB weights. Mosaic augmentation is disabled to preserve contiguous Bayer mosaicing patterns and localized physical noise distributions. Automatic Mixed Precision (AMP) is disabled to maintain the full 16-bit radiometric depth of the sensor measurements. Input resolution is 1024x864 pixels with a single-channel RAW input.

5.2 BEV-Space Perception Using BEVFormer

Unlike 2D detection, BEV-space perception demands rigorous geometric feature consistency across multi-view spatial projections. We utilize BEVFormer [Li et al., 2022], a transformer-based architecture that aggregates spatiotemporal features from multiple cameras into a unified Bird’s-Eye-View grid using spatial cross-attention.

A critical challenge in adapting BEVFormer for our dataset is that standard pinhole camera projection models fail under extreme fisheye distortion. To resolve this, we modified the spatial cross-attention (deformable attention) module to incorporate a 4th-order polynomial camera model, inspired by the WoodScape dataset geometry [Yogamani et al., 2019]. This custom geometric projection ensures that 3D reference points in the BEV space are accurately mapped to their corresponding highly distorted 2D pixel coordinates on the image plane.

Aside from adapting the spatial projection geometry, BEVFormer was modified to directly process single-channel 16-bit Bayer RAW TIFF images by changing only the first convolution layer of the ResNet-101 backbone from three to one input channel. Both RGB and RAW models were trained from scratch without pretrained weights using identical optimizer settings, training schedules, loss functions, and network architecture, comprising a ResNet-101 with DCNv2, an FPN, a BEV Encoder, and a DETR Decoder for predicting a 78×78 BEV grid of Vehicle and Pedestrian classes. Evaluation was performed using only the current frame, without temporal history, thereby isolating the influence of the sensor-domain representation.

6 Experimental Results

6.1 Image-Space Detection Resilience

Table 2 presents a quantitative comparison of YOLOv8 detection performance across ISP-processed RGB and real sensor-domain RAW representations and qualitative results of RAW and ISP processed images are shown in figure 2 and figure 3 respectively

Metric	ISP-Processed RGB	Real Sensor RAW
Vehicle Precision	0.84	0.79
Vehicle Recall	0.44	0.37
Vehicle F1-score	0.58	0.503
False Positives / Frame	0.95	1.08
mAP@0.50	0.525	0.513

Table 2: Quantitative comparison at conf=0.25 of YOLOv8 2D image-space detection across imaging domains.



Figure 2: Image-space detection inference using YOLOv8 in real 16-bit Bayer RAW automotive fisheye imagery.

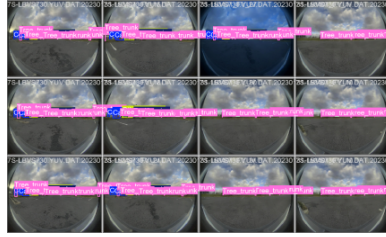


Figure 3: Image-space detection inference using YOLOv8 in ISP processed RGB automotive fisheye imagery.

The empirical evaluations demonstrate that sensor-domain RAW representations achieve semantic detection parity with conventional ISP pipelines. Specifically, RAW input yields a Vehicle F1-score of 0.503, close to the 0.58 achieved by the RGB baseline. Also the precision and recall for the RAW and RGB are close to each other. The RAW domain exhibits a robust $mAP@0.50=0.513$ which is remarkably closer to RGB $mAP@0.50=0.525$ and the resilient recall rates confirm that the fundamental semantic identity of objects is preserved prior to ISP intervention. The relatively low recall observed for both RGB and RAW models reflects the challenging nature of automotive fisheye object detection, characterized by severe radial distortion and large object-scale variation. Since both models were trained and evaluated under identical conditions, the comparison fairly isolates the effect of the input representation.

6.2 BEV-Space Geometric Stability

The most significant findings of this study emerge in the 3D BEV-space perception tasks. Table 3 summarizes the performance of the modified BEVFormer architecture operating under severe fisheye distortion and qualitative results of RAW and ISP processed images are shown in Figure 4



Figure 4: BEV-space detection inference using BevFormer in real 16-bit Bayer RAW and ISP processed Image in automotive fisheye imagery dataset. The left side shows inference on RAW image and right side shows inference on ISP processed image

Metric	ISP-Processed RGB	Real Sensor RAW
Vehicle Precision	0.726	0.726
Vehicle Recall	0.867	0.874
Pedestrian Precision	0.244	0.235
Pedestrian Recall	0.276	0.276
Average Orientation Error (AOE) Vehicle	0.040	0.041
Average Orientation Error (AOE) Pedestrian	0.394	0.417
Direct Corner Error Vehicle	0.599	0.581
Closest Corner Error Vehicle	0.583	0.566

Table 3: Comparison of BEVFormer @conf=0.3 multi-view 3D perception performance.

Despite the necessity for complex spatial projection and multi-view temporal feature aggregation, un-demosaiced RAW inputs achieve identical precision (0.726) and slightly superior recall (0.874) for vehicle detection compared to the RGB baseline. Furthermore, the geometric localization capabilities—measured via Average Orientation Error (AOE) and 3D corner errors—remain phenomenally stable. In fact, RAW representations marginally outperformed RGB in Direct Corner Error (0.581 vs. 0.599), indicating that the preservation of linear radiometric data and the absence of demosaicing artifacts actually aid the spatial cross-attention mechanisms in triangulating precise 3D geometric coordinates.

7 Discussion and Analysis of ISP Effects

The comparable, and in some metrics superior, performance of sensor-domain representations fundamentally challenges the necessity of hardware ISPs in autonomous perception pipelines. Traditional ISP operations—such as gamma compression and tone mapping—are mathematically non-linear transformations designed to map wide-dynamic-range scenes into a restricted, human-pleasing color space. While aesthetically beneficial, these operations structurally distort the linear relationship between scene radiance and pixel intensity.

Our findings demonstrate that state-of-the-art architectures equipped with deformable attention mechanisms can directly decode sparse, 16-bit physical sensor measurements. The profound robustness of RAW representations in 3D BEV space proves that the non-linear projection algorithms required to untangle severe radial fisheye distortion are mathematically compatible with raw Bayer sampling. By entirely bypassing the hardware ISP, autonomous driving systems can effectively eliminate the latency, power consumption, and hallucination artifacts associated with demosaicing, directly translating sensor-native measurements into reliable semantic and geometric intelligence.

8 Conclusion

This paper systematically investigates the paradigm of ISP-free autonomous perception, extending RAW-domain learning beyond rectilinear 2D detection into the rigorous domain of 3D Bird’s-Eye-View (BEV) spatial aggregation utilizing automotive fisheye cameras. By introducing the RAW-Det-Fisheye dataset, we facilitated a strictly controlled evaluation between 16-bit un-demosaiced sensor measurements and conventionally optimized RGB imagery.

Our comprehensive experiments using YOLOv8 and BEVFormer empirically establish that linear, sensor-domain representations preserve critical semantic structure and yield highly accurate 3D geometric localization, successfully navigating the compounded challenges of Bayer sparsity and severe radial distortion. This work strongly motivates a paradigm shift toward highly efficient, low-latency, sensor-native perception architectures jointly optimized for raw physics and machine intelligence, rather than human visual aesthetics. While our

current evaluation primarily focuses on the major dynamic object categories—such as vehicles and pedestrians—that are most critical for immediate crash prevention and automated braking systems, these promising baseline results open exciting avenues for future research. Moving forward, we plan to extend this ISP-free paradigm to traffic control elements, including traffic signs and traffic lights, as well as smaller, color-pattern-dependent objects like traffic cones and bollards. By demonstrating that core spatial and semantic intelligence is already robustly preserved in the RAW domain, this work lays a strong foundation for optimizing sensor-native networks across all edge cases, ultimately opening up the possibility of utilizing raw sensor data for full-scale, comprehensive autonomous driving applications.

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